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# CaseLinker: Interpretable ML Approaches for Analyzing Internet Crimes Against Children Reports

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*Technical Report Series #2* · github.com/mrinaalr/CaseLinker · Live Demo ·

*← Refined by Mrinaal. Black as written in March. Red: corrected. Yellow: right the first time. New pages at the back, R1 to R9.*

**Abstract**

This is the second in a monthly series of technical reports tracking CaseLinker's development as an open-source platform for cross-case analysis of Internet Crimes Against Children (ICAC) reports. Each report documents progress toward scaling from roughly 100 to thousands of publicly available case reports while integrating interpretable ML functionality, visualizations, and filtering tools that allow investigators to engage deeply with case patterns—without requiring repeated direct exposure to potentially disturbing material [2, 3]. This report covers: (1) integration of Named Entity Recognition (NER) via Stanza (Stanford NLP) into a hybrid deterministic-ML processing pipeline; (2) dataset expansion from 47 to ~~207~~ *10,711* cases across ~~seven source PDFs~~ *70 sources* spanning 2011–2024; and (3) early patterns emerging from the expanded dataset, including the distributed network of over ~~215~~ *4,238* law enforcement organizations spanning multiple states and years, along with recurring trends in platform usage and investigation structure across jurisdictions. Evaluation demonstrates 87.9% agency extraction coverage across the expanded dataset. As CaseLinker scales toward thousands of cases, integrating AI thoughtfully—in ways that remain auditable and that keep investigators in control—becomes both the central technical challenge and the most important design commitment. Automated insights and clustering will continue to improve, but they are not a substitute for the judgment of an experienced investigator or journalist working directly with this data. Expert intuition—the ability to recognize meaningful patterns from context and experience—remains something algorithms do not yet replicate [5]. CaseLinker is built to support that judgment.

*← 207 became 10,711. See R1.*

*Wrong yardstick. Don't count what you found. Count what you got wrong. R3. →*

**Keywords:** child exploitation, interpretable machine learning, deterministic extraction, cross-case analysis, analyst well-being, ICAC investigations

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# 1 Introduction

## 1.1 Monthly Report Series

The initial CaseLinker technical report [1] documented a working prototype evaluated on 47 publicly available AZICAC case reports from 2011–2014. That report established the five-layer architecture, the deterministic extraction pipeline, and the evaluation baseline. This series picks up from there, releasing a technical update each month that documents what was built, what was learned, and what the data is starting to show.

*Stopped at five. Wrote AfH. Now on Mechanics, with CaseLinker as the telescope. These reports document building it. Full notes are live on GitHub. →*

The goal of the series is not to report a finished system. It is to build one transparently—showing the reasoning behind each technical decision, the tradeoffs accepted, and document findings that become visible as the system grows. Each report is also a checkpoint: a record of what CaseLinker can and cannot do at a given scale, as the project evolves in studying and extracting meaningful patterns from this data.

## 1.2 What Changes at Scale

ICAC case reports contain attributes worth studying systematically: the nature of the offense (possession, production, extortion), the offender's age and relationship to the victim, which platforms or methods were involved, what investigative approach was used, whether the suspect was a repeat offender, a teacher, a family member, or a stranger, which agencies collaborated, how the case was prosecuted, and custom topics that can be defined by the researcher during analysis. The initial report demonstrated feasibility—structured extraction and cross-case clustering were demonstrably possible, and even that limited dataset surfaced patterns worth examining.

At ~~207~~ *10,711* cases spanning 2011–2024, more comes into focus. Law enforcement agency involvement becomes visible at scale: ~~**215**~~ *4,238* **unique organizations** appear across ~~207~~ *10,711* cases, with ~~**171 (79.5%)**~~ *2,337 (55.1%)* appearing in only one case—demonstrating a wide, distributed network of local agencies converging around a stable institutional core of federal partners and NCMEC. ICAC task forces appear in ~~**51.2%**~~ *36.3%* of cases, the FBI in ~~**9.2%**~~ *18.2%*, ~~with its~~ *and* specialized Child Sexual Exploitation Unit*s* ~~accounting for roughly half of those appearances~~ *in 4.0% of cases*. That federal-local structure is not obvious from any single case; it only emerges in aggregate.

*Wrong. Unknown is not stranger. The 91.8% counted both. Explicit strangers: 8 cases. Q4. →*

Other patterns are equally instructive. ~~Stranger perpetrators account for 91.8% of cases—a figure that cuts against common assumptions about who commits these crimes.~~ *A relationship to the victim is recorded as someone the victim knows, such as family, a teacher, or a coach, in 7.3% of cases and as a stranger in 0.1%; the remaining 92.6% record none.* Possession dominates over production (~~66.2%~~ *65.8%* versus ~~2.9%~~ *11.0%*), ~~yet~~*;* ~~37.7%~~ *52.9%* of cases ~~involve~~ *mention* sexual assault, and ~~6.8%~~ *3.0%* involve victims under 12. The average age gap between reported perpetrator and victim age is ~~22.6 years~~ *27.1 years*. Platform data is present in ~~only 32.4%~~ *60.0%* of cases, ~~but~~ *and* where it appears, online methods (~~21.3%~~ *31.9%*), chat platforms (~~8.2%~~ *7.7%*), and Facebook (~~4.8%~~ *3.6%*) lead—with Snapchat and Discord beginning to appear in more recent cases.

*← All stats here are estimates and lower bounds. Reproduce with Q3 to Q8.*

These are early signals and informal statistics easily derived from this database, not conclusions. The research questions they open will scale along-

side the dataset—and more will emerge as it grows, including: how do federal-local coordination structure vary by offense type or geography? Do proactive versus reactive investigation split predict prosecution outcomes? How have platform patterns shifted between 2011–2014 and 2022–2024, and do newer platforms correlate with different severity profiles? Are there common investigative methods that correlate with better outcomes for the youngest and most severely harmed victims? The infrastructure to find answers to these and more questions in this landscape is being built.

*← Built. Ask the graph in SPARQL, or hand an agent the tools. R6, R7.*

This is publicly available data containing no personally identifiable information, analyzed under a Not Human Subjects Research determination (HRPO #7668) from the University of Massachusetts Amherst Human Research Protection Office, confirming the research contains no private or identifiable information under federal regulations [45 CFR 46.102(f)(1), (2)].

# 2 Design Philosophy and the Role of Interpretable ML

## 2.1 The Problem with Scaling Without ML

The initial system used exclusively deterministic, regex-based extraction. This was the right choice for a 47-case prototype: interpretable, auditable, no training data required. But regex does not scale gracefully to an environment with hundreds of agencies, inconsistent reporting formats, and 13 years of evolving language. The cost of maintaining exhaustive pattern lists grows with every new source PDF added. More importantly, patterns you did not anticipate do not get extracted at all—they simply disappear from the data.

*← True, and NER was not the cure. A model that fills gaps still lets one method outrank the rest. See R2.*

The 207-case dataset makes this concrete. Before NER integration, agency extraction was limited to patterns explicitly encoded in the regex library. After integration, the system identified 215 unique law enforcement organizations—dramatic increase over the limited regex-only baseline—by recognizing organizational entities from linguistic context rather than direct matching alone.

## 2.2 What NER Unlocks

*Keep Stanza. Once per PDF, confidence capped at 0.65. It may agree. It may not decide. R2. →*

Named Entity Recognition using Stanza (Stanford NLP) adds a semantic layer to extraction. Where regex asks "does this text match a known pattern," NER asks "what kind of thing is this text referring to." For a dataset covering 61 ICAC task forces, federal agencies, county prosecutors, state police, and international partners, the difference is significant.

The extraction pipeline processes four entity types: `ORG` (filtered to law enforcement agencies), `DATE` (separating ages and event dates using contextual heuristics), `LOC`/`GPE` (geographic entities), and `PER` (excluded to avoid extracting reporter names). The result is a richer, more complete data that we can capture and analyze.

## 2.3 Patterns Immediately Discovered in the Expanded Dataset

With ~~207~~ *10,711* cases now processed, several patterns have emerged that were not visible at smaller scale. These are observations from the current

dataset, not conclusions—the sample remains limited and some figures will shift as coverage grows.

*4,238 after merging name variants and cutting fragments. See* `/api/stats`. →

**The collaborative network.** The ~~215~~ *4,238* unique organizations span a wide distribution: ~~**171 (79.5%)**~~ *2,337 (55.1%)* appear in only one case, while ~~44~~ *1,901* appear in multiple cases. The most frequently recurring are ICAC Task Forces (~~106 cases, 51.2%~~ *3,888 cases, 36.3%*), NCMEC (~~48 cases, 23.3%~~ *3,240 cases, 30.2%*), and the FBI (~~19 cases, 9.2%~~ *1,954 cases, 18.2%*), including its specialized Child Sexual Exploitation Unit (10 cases). A structural detail worth noting: despite comprising only ~~~10~~ *124* unique organizations, federal agencies account for ~~~80~~ *14,277* total case appearances—comparable to the ~~~375~~ *19,556* case appearances from ~~~195~~ *4,114* unique local agencies. Federal partners appear less often in count but recur heavily across cases, with an average of ~~~8.5~~ *115.1* cases per federal agency compared to ~~~2~~ *4.8* cases per local agency.

← *Local includes state agencies and ICAC task forces. Open any agency's IRI to browse its cases. R6.*

**Multi-agency coordination.** The ~~44~~ *1,901* organizations appearing in multiple cases form a recurring ~~institutional core~~ *layer*, but the majority of the network is non-recurring. That ~~79.5%~~ *55.1%* ~~long tail~~ *majority* of single-appearance agencies suggests each investigation draws on a highly localized set of partners—a pattern that may demonstrate how ICAC task forces operate in practice, but one that has not previously been quantifiable from public records alone.

**Geographic and temporal distribution.** Cases span multiple states and 13 years. Platform data is ~~sparse—present in only 32.4% of cases—but~~ *present in 60.0% of cases, and* where it appears, the mix shifts noticeably between 2011–2014 and 2022–2024, with Snapchat and Discord appearing in more recent cases alongside the persistent dominance of generic online and chat references. Quantifying this shift systematically is a goal of Report #3.

**Investigation structure.** The investigation type is unknown in ~~46.9%~~ *56.8%* of cases—not a data error, but a reflection of inconsistent documentation in public summaries. Of the cases where type is recorded, proactive investigations (~~6.3%~~ *0.8%*) outnumber reactive ones (~~1.0%~~ *two cases*), though the sample is too small to draw reliable conclusions about outcome differences. This remains an open analytical question as the dataset grows.




Figure 1: Initial exploratory analysis from the CaseLinker dataset. These figures illustrate feature extraction coverage, law enforcement organization participation patterns, and platform usage trends across ICAC case reports spanning 2011–2024.

*← Only 207 cases in these plots. Leave them. They are the first look.*

## 2.4 Interpretability as a Non-Negotiable

Adding ML to CaseLinker does not mean accepting opaque black-box outputs. Every extraction is subject to the same auditability requirement as the deterministic pipeline: an analyst must be able to trace any feature back to its source text and understand how it was identified. For NER specifically, this means the merge logic is explicit and inspectable ~~— pattern-based extraction takes precedence in conflicts~~, and NER contributions are flagged as such in the database. The goal is a system where an analyst can always ask "why does this case show this agency" and get a straightforward answer.

*← Crossed out. No method outranks another now. Let them vote, by evidence family. R2.*

*The "why" has an answer now: a character span, a method, a confidence. R5. →*

A simpler alternative—uploading case narratives directly to a commercial large language model—is not viable for this domain. Beyond the sensitivity of the content itself, commercial LLM APIs may temporarily retain submitted inputs and outputs for abuse monitoring and safety purposes, under data-usage policies that vary by provider [4]. Outputs are probabilistic and cannot be traced to source text [8], violating the auditability requirement for legal and research contexts. There is no established chain of custody for LLM-derived extractions, and models may reproduce or surface sensitive details in unpredictable ways. CaseLinker's architecture exists precisely because the standard shortcuts are not appropriate here.

*← True for extraction. Let a small local model testify on the leftovers, under oath. Hand agents bounded tools. R4, R7.*

This matters not just for analyst trust but for legal and reporting contexts. Case analysis that informs investigative decisions needs to be explainable. A feature or statistic extracted by an opaque model that cannot be traced to source text is not useful in that context—it is a liability.

## 2.5 Future ML Integrations

The ML roadmap for CaseLinker follows the same principle as the NER integration: each addition must be optional, independently verified, and clearly separated from the deterministic core. Three capabilities are under active development.

*None built. The time went to extraction, provenance and the graph. R9. →*

**Semantic similarity.** Current case clustering uses weighted Jaccard similarity to determine sub-groups across 5 cluster topics. Sentence transformer embeddings (`all-mpnet-base-v2`) would enable similarity based on meaning rather than keyword overlap—finding cases that describe the same offender methodology even when the surface language differs. For investigative narratives with legal terminology, a domain-adapted model such as nlpaueb/legal-bert-base-uncased [7] will be evaluated against the general-purpose baseline before any production integration.

**Enhanced grouping.** Semantic embeddings also make it possible to group cases and create diverse new clusters that traditional keyword methods miss—for example, similar platform usage patterns described differently across task forces or consistent investigative approaches that do not share exact terminology. Entity networks derived from NER output could additionally visualize relationships between organizations, platforms, and case characteristics across the full dataset.

*← Got there another way: case graphs, with agencies, locations and platforms as nodes. R6.*

**Content sanitization.** Summarization models (BART/T5) may eventually generate analytical case summaries that preserve investigative facts while omitting explicit detail—reducing direct exposure to disturbing material during review. This capability requires careful evaluation: a summary that loses legally relevant detail is worse than no summary. It is listed here as a research direction, not a planned release.

All ML dependencies are maintained in a separate `requirements-ml.txt` and are not required for core system operation. The system reverts gracefully if ML components are unavailable, ensuring the deterministic pipeline remains the reliable foundation.

*Only the local LLM is optional now. Off unless you switch it on. →*

# 3 Audit and Performance

## 3.1 Verification Before Storage

All ML-extracted features are merged with deterministic outputs and validated before being written to the database. The merge logic enforces explicit rules: ~~pattern-based extraction takes priority for fields where regex achieves high precision~~, and NER fills gaps rather than overwriting confident extractions. Normalization handles surface-form variations before any feature reaches storage—apostrophe inconsistencies, spacing artifacts, and known abbreviation errors (e.g., `ZICAC` → `AZICAC`) are resolved at the merge layer, not post-hoc.

*← Don't merge. Resolve. Span gate first, then the vote. Doubtful mentions go to review. R2.*

The result is a database where every stored feature has passed through an explicit, inspectable processing chain. There are no silent failures and no opaque model outputs stored without a corresponding source.

*Enforced now. A wrong final is worse than a mention left in review. →*

## 3.2 Interactive Audit Interface

CaseLinker's audit interface allows case-by-case review of all extracted features alongside the original case text. Hovering over any extracted feature highlights the corresponding source passage, making the extraction traceable without requiring access to the underlying code. This is available for every case in the database and covers the full feature schema: perpetrator and victim demographics, platforms, agencies, investigation type, severity indicators, charges, and evidence volume.

*← Better: store the span itself. Every claim, character by character. R5.*

The audit interface serves two purposes. For research use, it enables verification that the extraction pipeline is behaving correctly as new source PDFs are added. For any future practitioner use, it provides the transparency required to trust automated outputs—an analyst can spot-check any case, confirm any feature, and identify where extraction falls short.

## 3.3 Performance

NER processing runs during ingestion, not at query time. All clustering, triage, and insight generation are precomputed and stored, so production query performance is unaffected by the ML layer. Current benchmarks on the 207-case dataset:

*10.8 hours to extract all 10,711. Minutes to re-vote. Never re-extract for a rule change. R2. →*

Table 1: Processing Performance (207 cases, 7 PDFs)

| Operation | Time |
|---|---|
| Full ingestion with NER (7 PDFs) | 30–60 seconds |
| Full analysis pipeline | 2–5 seconds |
| API endpoint queries | <100ms |

Projected at 2,000 cases, ~~ingestion is estimated at 5–10 minutes~~ (one-time cost per batch), with query performance maintained via precomputed clusters.

*← Optimistic. Four layers cost more than one. Worth every hour.*

# 4 Moving Forward

What this dataset is starting to show is that child exploitation investigations are not isolated events handled by a single agency. They are distributed,

collaborative, and patterned—across states, years, platforms, and organizations. That structure is only visible in aggregate, and I am very interested in studying it.

CaseLinker is the primary vessel for this motivation. Not to replace investigative judgment and time spent reading cases, but to provide analysis tools and a platform to make answering key, challenging questions as easy as possible.

The (fun) challenge is that scaling this responsibly takes time. Each new source PDF requires validation. Each new ML component requires evaluation against ground truth. The audit interface exists precisely because these checks need to be visible and ongoing, not assumed. The monthly report series is part of that commitment—a public record of what the system can do at each stage, and what it cannot yet.

*No gold set. Does the stored value match its source, in context? Two frozen samples. Keep one untouched. R3. →*

## 5 Roadmap

- **3 months (Report #4, June 2026).** Target: 1,000+ cases. The NCMEC CyberTipline annual reports alone contain **roughly 2,800 publicly available cases** across 2022–2024, of which 160 are currently ingested. Analyzing this existing source and additional reports from at least one other task force beyond AZICAC will bring the dataset past the 1,000-case threshold without requiring any new data acquisition. Primary engineering focus: extending the ingestion layer to handle formatting variation across task forces. ML focus: temporal trend analysis across the full 2011–2024 span, quantifying platform evolution and investigation method shifts. Visualization: platform will remain responsive and visually accessible, with multiple avenues for analysis as the dataset grows.

*← Overshot tenfold. 10,711, from 70 sources.*

- **6 months (Report #7, September 2026).** Target: 2,000+ cases. Introduce sentence transformer embeddings for semantic case similarity—opt-in, clearly separated from the deterministic core, evaluated against the weighted Jaccard baseline. Begin severity classification experiments using the labeled severity indicators already in the dataset as ground truth. If institutional partnerships and user studies are desired, integrate feedback from practitioner review.

*Not built. Extraction first. Embeddings can wait. →*

- **12 months (Report #13, March 2027).** Target: Comprehensive coverage of publicly available ICAC reports across all a wide array of online sexual exploitation instances. Full temporal analysis across all available years. Network analysis for multi-agency and multi-jurisdiction case connections. Peer-reviewed publication submission documenting the system and findings. Active user study with researchers or analysts if institutional partnerships allow.

*← Groundwork done: the case graphs and the audit loop. R6, R8.*

## 6 Conclusion

Version 2 of CaseLinker advances the system on two fronts. The hybrid NER-deterministic pipeline achieves 87.9% agency extraction coverage on a 207-case dataset spanning 13 years and two source types, directly addressing the coverage limitation of the regex-only baseline. More importantly, the expanded

dataset has started to reveal insights: a distributed collaborative network of ~~215~~ *4,238* law enforcement organizations, ~~80%~~ *55.1%* of which appear in only one case—a pattern that is invisible without cross-case analysis and that has real implications for understanding how these investigations are conducted.

The deeper goal of this project is not technical metrics. It is giving investigators and researchers tools that let them understand the landscape of child exploitation at scale—platform trends, offender patterns, investigation strategies, jurisdictional networks—without requiring them to carry the psychological weight of repeated direct exposure to case material. Every technical decision in CaseLinker is evaluated against that standard: does it help people go further, while protecting them from going through more?

*Add a second test: does it show its work?* →

Report #3 will document progress toward 1,000 cases and introduce temporal trend analysis across the 2011–2024 dataset.

*CaseLinker is available at github.com/mrinaalr/CaseLinker released under the MIT License.*
*Live demo:* ~~*web-production-13a2.up.railway.app*~~ *caselinker.up.railway.app.*

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# R1 What Changed Since March

Seven months changed CaseLinker more than this report predicted, in three ways. CaseLinker reads more: the corpus grew from 207 cases to 10,711, drawn from 70 sources. CaseLinker decides differently: regex plus NER became four independent layers, one of them an optional local model, voting and abstaining when the evidence does not support a value. And CaseLinker shows its work to people and agents alike: every value carries its provenance, graphs are validated against the ontology with SHACL, entities are grounded to public authorities, names resolve, and 54 tools expose all of it to agents over the Model Context Protocol [16].

Table 2: CaseLinker in March and in October 2026.

| | **March 2026** | **October 2026** |
|---|---|---|
| Cases | 207 | 10,711 |
| Sources | 7 PDFs | 70 sources, several international |
| Agencies | 215 | 4,238 |
| Extracted features | — | 131,332 |
| Extraction | Regex plus Stanza NER; pattern precedence on conflict | Four layers plus an optional local LLM; no precedence, a vote by evidence family |
| Evidence per value | Audit interface: hover to a passage | Character span, method, version and confidence, stored as a provenance graph |
| Identifiers | Database rows | Permanent IRIs that resolve |
| Graphs | None | Two per case, the case and its provenance, typed with CAC, UCO and CASE |
| Validation | Spot checks in the audit interface | SHACL shapes on every provenance graph |
| Entity names | Raw strings | Grounded to Census and FBI references |
| Agent access | None | 54 MCP tools |
| Re-running a rule change | Full re-extraction | Re-vote in minutes |

October delivers scale, evidence for every value, and tools that agents can call, while remaining true to the goals and constraints the domain motivated: investigators stay in control, every output traces to its source, and analysts can do the work without repeated direct exposure to the material.

# R2 Four Layers, One Vote

Section 2.1 ended on a worry: patterns I did not anticipate would never be extracted. NER was the first answer, and it was not enough, for a structural reason. A merge rule that lets one method take precedence means one reader always wins the argument. Phase 3 replaced precedence with testimony.

**Four layers read every case.** Each writes its own claim and none can overrule another:

1. *Surface patterns*: regex, spaCy token patterns, an age gate, and compiled rules.

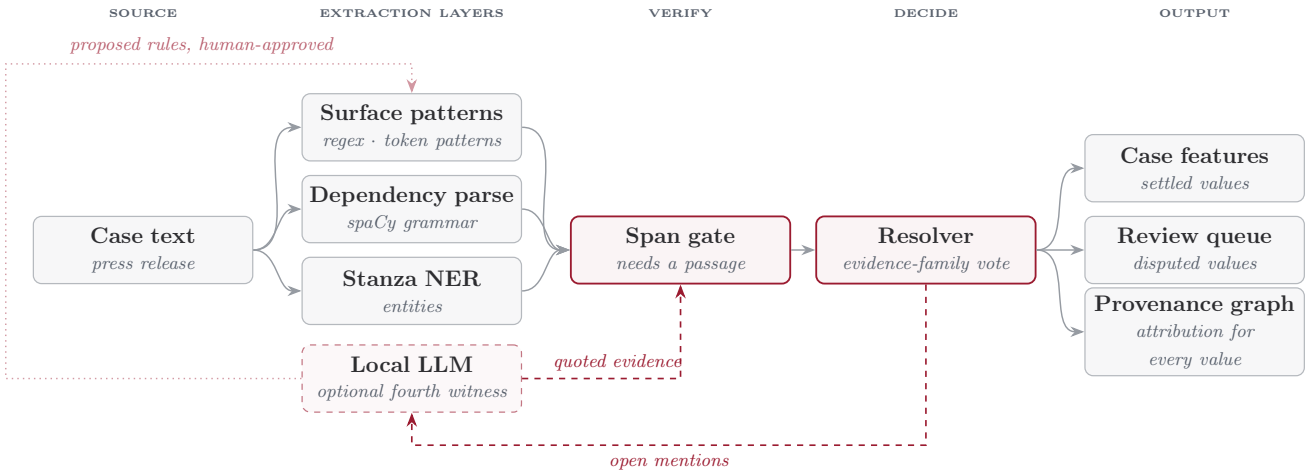



Figure 2: The Phase 3 pipeline. The span gate and the resolver (red) are the two steps every value must pass. The local model reads only what the deterministic layers left open, and its quotes go back through the same gate.

2. *Grammar*: a spaCy dependency parse ties each age to the person it describes. A role comes from the sentence, never from the size of a number.
3. *Entities*: Stanza NER, once per PDF [6].
4. *An optional local model*, described in R4.

Statute lookup, temporal tags, and human overrides sit beside them as specialists.

**Every claim is an assertion**: a value, a role, a confidence, the method and its version, and the character span in the original text. Assertions are appended and never edited. A span gate rejects any value the passage does not support, and any age or sentence that is not in its expected form.

**Votes count by evidence family, not by reader**, so two regexes are one vote. An unknown role is an abstention. A family that contradicts itself abstains. Two families that disagree send the mention to review. Two families that agree settle it. One family can settle a value alone only at confidence 0.7 or above, with no other family disagreeing. The Stanza name cue sits at 0.65, so entity recognition can corroborate a value and cannot finalize one.

The standard is simple: a wrong final is worse than a mention left in review. The cost is visible. On the full corpus, 73.7% of age finals rest on two independent families in agreement, 19.0% of age mentions remain in review, and 24.1% of cases carry at least one open mention.

Resolution only reads assertions. A new rule set can therefore re-vote the whole corpus in minutes without re-extracting, against 10.8 hours for a full re-extraction of 10,711 cases.

## R3 How the Numbers Are Measured

I did not build a gold set, and I do not want one. The standard is whether a stored value matches its source passage and makes sense in context. An age is wrong if it belongs to a decoy rather than the person it was stored for. A sentence is wrong if it is a maximum penalty, a prior case's term, or CyberTip boilerplate standing in for the sentence imposed.

Two samples of 200 paired case ids carry the evidence. The *development* sample (seed 20261017) was used to find and fix errors. The *holdout* (seed 20261018, excluding the development ids) was judged after every fix and produced no changes. Both give one seat

to each of the 63 sources present in both versions and fill the rest by source size. The same ids are scored under the 2.x pipeline and under Phase 3. Every verdict was human-agent reviewed against the source passage, and the ids and per-case verdicts are in the repository.

Table 3: Error rate by sample and field, with 95% Wilson intervals [10].

| Sample | Field | Pipeline | Wrong/stored | Rate | 95% interval |
|---|---|---|---|---|---|
| Development | Ages | 2.x | 0/177 | 0.0% | [0.0, 2.1] |
| | | Phase 3 | 0/179 | 0.0% | [0.0, 2.1] |
| | Sentences | 2.x | 33/249 | 13.3% | [9.6, 18.0] |
| | | Phase 3 | 0/95 | 0.0% | [0.0, 3.9] |
| Holdout | Ages | 2.x | 1/172 | 0.6% | [0.1, 3.2] |
| | | Phase 3 | 2/201 | 1.0% | [0.3, 3.6] |
| | Sentences | 2.x | 55/281 | 19.6% | [15.4, 24.6] |
| | | Phase 3 | 1/118 | 0.8% | [0.2, 4.6] |

On the holdout, Phase 3 cuts the sentence error rate from 19.6% to 0.8% and stores 17% more ages at comparable accuracy: 1.0% against 0.6%, a one-error difference. The intervals separate cleanly on sentences. The Phase 3 upper bound, 4.6%, sits well below the 2.x lower bound, 15.4%.

Zero errors is not a point estimate. The development result of 0 in 95 sentences still has an interval that reaches 3.9%, and the development sample is the one the fixes were tuned on. The holdout is the number to quote.

## R4 The Model Under Oath

Section 2.4 argued that sending case narratives to a commercial language model is not viable here. That argument stands for extraction, and the local model works within the same constraints.

The model runs on the investigator's machine (Ollama, `qwen3:8b`) and is off unless switched on. It reads sentence windows, never whole releases, and only the mentions the deterministic layers could not settle. It returns a role from a fixed list together with a quote from its window. A quote that fails the span gate is dropped. Its claims carry confidence 0.6, below the 0.7 needed to finalize alone, so the model can confirm or veto a value and can never store one by itself. Replies are cached by model digest, prompt, and parameters. The Phase 3 pipeline is fully reproducible: the LLM is a witness under oath, with one passage, one answer from a fixed list, and the same testimony every time it's asked.

**A first measurement.** I drew 200 of the 3,190 cases that had an open mention (seed 20261019) and ran the model over them on a copy of the corpus. It added 14 finals and vetoed none; 25 of its quotes failed the span gate. Reading the 14 against their sources, 12 were right. The two errors were ages from missing-child recovery roundups, where one page holds many unrelated cases. This is one draw and one model, judged by reading, so I report it as exploratory. It does show what the model is for: it recovers the child or the defendant the sentence is about, and it has not yet overturned a deterministic final.

**Rules, not models.** What a model learns should end as auditable code. A drafting script turns reviewer-confirmed quotes into a candidate regex, and nothing runs until a person approves it. No rule has been approved yet.

## R5 Provenance and the Ontology

Every case becomes two graphs. The first is the case itself, typed with the CAC ontology alongside UCO and CASE [15, 14]: an investigation, offender and victim roles, age estimates. The second is its provenance.

In the provenance graph every assertion is a W3C Web Annotation [11] and a CAC `AssessmentResult`. It carries a `TextPositionSelector` (characters 7,081 to 7,083), a `TextQuoteSelector` (the exact text, "22"), a confidence, a method, and the time it was issued. Extraction runs and methods are PROV-O activities and software agents [12], and each field resolution records the assertions it was derived from. Three terms are new, under `w3id.org/caselinker/ns/provenance#`: `evidenceFamily`, `resolutionStatus`, and `ruleSetVersion`.

Table 4: Three readers, one span: the age 22 in case `nj_ag_2017_001`, characters 7,081 to 7,083.

| Method | Family | Role claimed | Confidence |
|---|---|---|---|
| `regex` | surface pattern | perpetrator | 0.90 |
| `spacy_dep` | dependency | perpetrator | 0.85 |
| `ner_stanza` | entities | none (abstains) | 0.50 |

Two independent families assign the same role. The entity layer finds the age and makes no role claim, which counts as an abstention and not as a vote against. Nothing here is hidden: each row is a node a person or a query can open.

SHACL shapes [13] validate every provenance graph. A graph that lacks a rule-set version, an evidence family, or an annotation's quote, span, confidence, or issue time does not conform. The CAC shapes are pinned in the repository, and validation fails loudly when shapes cannot load. A graph that cannot say where a value came from does not validate.

The March audit interface let me hover over a feature and see a passage. This is the same idea stored as data. Any tool that reads Web Annotation can find the span, and any SPARQL query can walk from a case to its quoted evidence in one hop.

## R6 Names That Resolve

Cases, agencies, platforms, and locations are named under a permanent namespace registered at `w3id.org/caselinker` [17]. A case is `.../resource/case/{case_id}`, and its provenance graph sits at the same name with `/provenance` appended. The redirect keeps the path, so a case's permanent name lands on that case. If the host moves, only the redirect changes; the IRIs remain stable.

A name should answer with the thing it names. Open a case and the application returns the case: its type, its agencies and locations, a link to its provenance. A browser receives a readable page. A program that asks for Turtle or JSON-LD receives the graph. The vocabulary terms resolve to their own definitions. Agency, platform, and location pages carry a section that lists every case referencing them, so the graph can be walked from a location to the cases that mention it, and onward.

Case ids are minted at ingest and are stable within a release. A case is cited by the DOI of the release it belongs to.

## R7 Agents Over the Data

The March report said analysis tools should make hard questions easy to ask. The Model Context Protocol [16] allows analysts to team with agents to ask hard, cross-case questions of the whole corpus and check every answer against its source. CaseLinker's server exposes 54 typed tools in five groups: corpus reads (counts, facets, case pages, cohorts); graph reads (neighbors, cohort comparison, on-demand case graphs); triage; provenance (`get_case_assertions` returns every reader's claim for a case, `get_field_disagreements` returns every open disagreement); and collectors for public records (DOJ press releases, CourtListener and RECAP filings). A single `sparql_query` tool reaches the graph itself.

The tools are bounded by design. `sparql_query` refuses updates and federated `SERVICE` calls, applies a default row limit of 1,000 and a 90-second timeout, and can target the live endpoint, a local Oxigraph store, or a local N-Quads file. The hosted server registers 44 of the 54 tools: the eight collectors that write and two local-disk duplicate checks run only on a local server. Six sensitive tools (bulk exports, full case records, assertions, and the SQL assistant) sit behind a trusted key.

This keeps the March commitment intact. An agent that works through these tools inherits the audit trail: every number it quotes can be asked again, and every disagreement it meets is visible. In practice agents do three jobs: they collect records, analyze cohorts, and audit graphs, presenting patterns with the evidence behind each case, for the analyst to reason over.

## R8 The Graph as an Instrument

Phase 3 made every value traceable. It did not make every value sensible. Nothing in the span gate or the vote notices that one county cannot sit in three states, and the graph noticed what the extractors could not. Counting locations by the number of cases that reference them, I found that of 6,938 distinct location names in the corpus, 1,561 (22%) were shared across states. "Cumberland County" stood for counties in New Jersey, North Carolina, and Pennsylvania at once, and "the Southern District" for several federal districts.

The fix is grounding: link each mention to an authority, so that one real location gets one name. Locations resolve to the U.S. Census gazetteer [18], and agencies to the FBI Crime Data Explorer registry [19] (19,646 agencies, each with an ORI code) plus a short table of federal agencies. Both references are pinned snapshots with checksums, and the pipeline fails loudly if either is missing. Grounding runs after extraction and resolution, from stored data, so it re-runs without re-extracting. Each grounding records the reference version, the matched code, the state hint, and where the hint came from. A mention that cannot be placed stays ambiguous. International locations are out of scope for a U.S. gazetteer and are counted separately.

**Direction.** The extractors measure, and the resolver averages the measurements. What is missing is conservation laws: facts a measurement cannot violate, such as one person holding one age, or a local agency acting near a location in its case. Written over the ontology and run as queries against the graph, such checks would flag impossible shapes. Every triple already carries its provenance, so a violation points to the exact assertions that cannot all be true. The next step is a loop in which an agent drafts a deterministic rule from a cluster of violations, and a rule ships only if it removes errors without adding new ones [20].

## R9 What Was Promised, and What Happened

Table 5: The March promises, scored in October.

| **Promise (March 2026)** | **October 2026** |
|---|---|
| 1,000+ cases by June | 10,711 cases, 70 sources |
| Ingestion across task-force formats | Done, including international sources and a gate against storing a URL twice |
| Entity network of agencies and locations | Shipped as case graphs; agency and location pages list their cases |
| Audit by hovering to a passage | Superseded: every claim stored with its span (R5) |
| Ingestion in 5–10 minutes at 2,000 cases | Not met: 10.8 hours for a full re-extraction; re-voting takes minutes |
| Sentence-transformer similarity | Not built |
| Legal-domain embeddings | Not run |
| Summarization for sanitization | Not built |
| Peer-reviewed submission | Ahead |

**Data and credit.** The sources are public and redacted: ICAC reports, NCMEC materials, DOJ press releases, and federal court filings. No personally identifiable information was processed, and the work was done in accordance with HRPO determination #7668. The age-extraction validation sampling came from Uriel Tulinabo, and the additive source provenance that the Phase 3 provenance graphs extend came from Forest Savage. CaseLinker is released under the MIT License at github.com/mrinaalr/CaseLinker, with the live system at caselinker.up.railway.app and the Phase 3 release archived on Zenodo [9].

## References for the revised pages

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# A Reproducing the Statistics

Q1 to Q9 reproduce the figures in the revised pages that the graph can express, and Q10 to Q12 walk the graph. Run any of them against the CaseLinker SPARQL endpoint, through the `sparql_query` tool, or on a local Oxigraph loaded from the release N-Quads. Each case is one named graph with a companion provenance graph, and Q1 to Q9 count case graphs, so the denominator is Q1. All queries assume the prefixes in the first listing. The count of distinct law-enforcement organizations is the `unique_organizations` field of `GET /api/stats`. The agency recurrence and the federal and local figures are computed from the database after the same normalization, because one agency can carry several IRIs, which is why Q2 returns a higher number. Other figures the graph does not carry come from the database, which is not public: the sexual-assault flag, generic online platform references, the reactive-investigation count, and the platform presence, chat, and Facebook shares, for which Q8 returns the lower graph counts. The FBI Child Sexual Exploitation Unit count comes from case text, because agency labels fold the unit into FBI: ten releases name it (azicac_2012_007 to azicac_2012_011, azicac_2013_002 and azicac_2013_003, azicac_2014_001 to azicac_2014_003).

**Prefixes** [prepend to every query]

```
PREFIX cac:   <https://cacontology.projectvic.org#>
PREFIX core:  <https://cacontology.projectvic.org/core#>
PREFIX mj:    <https://cacontology.projectvic.org/multi-jurisdiction#>
PREFIX tf:    <https://cacontology.projectvic.org/taskforce#>
PREFIX det:   <https://cacontology.projectvic.org/detection#>
PREFIX prod:  <https://cacontology.projectvic.org/production#>
PREFIX cust:  <https://cacontology.projectvic.org/custodial#>
PREFIX oa:    <http://www.w3.org/ns/oa#>
PREFIX prov:  <http://www.w3.org/ns/prov#>
PREFIX dcterms: <http://purl.org/dc/terms/>
PREFIX clprov: <https://w3id.org/caselinker/ns/provenance#>
```

**Q1** [Cases]

```
SELECT (COUNT(DISTINCT ?g) AS ?cases) WHERE {
  GRAPH ?g { ?case a cac:CACInvestigation }
  FILTER(STRSTARTS(STR(?g), "https://w3id.org/caselinker/resource/case/")
         && !STRENDS(STR(?g), "/provenance"))
}
```

**Q2** [Agency IRIs (an upper bound on distinct agencies; one agency can carry several IRIs)]

```
SELECT (COUNT(DISTINCT ?agency) AS ?agencies) WHERE {
  GRAPH ?g { ?case mj:involvesAgency ?agency }
  FILTER(STRSTARTS(STR(?g), "https://w3id.org/caselinker/resource/case/")
         && !STRENDS(STR(?g), "/provenance"))
}
```

**Q3** [Share of cases involving ICAC task forces, NCMEC, and the FBI]

```
SELECT ?agency (COUNT(DISTINCT ?g) AS ?n) ?cases
       (ROUND(1000 * COUNT(DISTINCT ?g) / ?cases) / 10 AS ?pct) WHERE {
  { GRAPH ?g { ?case mj:involvesAgency ?a . ?a a tf:StateICACtaskForce }
    BIND("ICAC task forces" AS ?agency) }
  UNION
  { GRAPH ?g { ?case mj:involvesAgency <https://w3id.org/caselinker/resource/agency/ncmec> }
    BIND("NCMEC" AS ?agency) }
  UNION
  { GRAPH ?g { ?case mj:involvesAgency <https://w3id.org/caselinker/resource/agency/fbi> }
    BIND("FBI" AS ?agency) }
  FILTER(STRSTARTS(STR(?g), "https://w3id.org/caselinker/resource/case/")
         && !STRENDS(STR(?g), "/provenance"))
  { SELECT (COUNT(DISTINCT ?g2) AS ?cases) WHERE {
      GRAPH ?g2 { ?c a cac:CACInvestigation }
      FILTER(STRSTARTS(STR(?g2), "https://w3id.org/caselinker/resource/case/")
         && !STRENDS(STR(?g2), "/provenance"))
  } }
} GROUP BY ?agency ?cases
```

**Q4** [Cases by victim relationship: known person, stranger, none recorded]

```
SELECT ?known ?stranger (?cases - ?known - ?stranger AS ?none_recorded) ?cases
       (ROUND(1000 * ?known / ?cases) / 10 AS ?pct_known)
       (ROUND(1000 * ?stranger / ?cases) / 10 AS ?pct_stranger) WHERE {
  { SELECT (COUNT(DISTINCT ?g) AS ?known) WHERE {
      GRAPH ?g { ?rel cust:relationshipType ?type . FILTER(?type != "stranger") }
      FILTER(STRSTARTS(STR(?g), "https://w3id.org/caselinker/resource/case/")
         && !STRENDS(STR(?g), "/provenance"))
  } }
  { SELECT (COUNT(DISTINCT ?g) AS ?stranger) WHERE {
      GRAPH ?g { ?rel cust:relationshipType "stranger" }
      FILTER(STRSTARTS(STR(?g), "https://w3id.org/caselinker/resource/case/")
         && !STRENDS(STR(?g), "/provenance"))
  } }
  { SELECT (COUNT(DISTINCT ?g2) AS ?cases) WHERE {
      GRAPH ?g2 { ?c a cac:CACInvestigation }
      FILTER(STRSTARTS(STR(?g2), "https://w3id.org/caselinker/resource/case/")
         && !STRENDS(STR(?g2), "/provenance"))
  } }
}
```

**Q5** [Cases with a possession or production event]

```
SELECT ?event (COUNT(DISTINCT ?g) AS ?n) ?cases
       (ROUND(1000 * COUNT(DISTINCT ?g) / ?cases) / 10 AS ?pct) WHERE {
  VALUES (?event ?suffix) { ("possession" "/event/possession")
                            ("production" "/event/production") }
  GRAPH ?g { ?case cac:hasStep ?step }
  FILTER(STRENDS(STR(?step), ?suffix))
  FILTER(STRSTARTS(STR(?g), "https://w3id.org/caselinker/resource/case/")
         && !STRENDS(STR(?g), "/provenance"))
  { SELECT (COUNT(DISTINCT ?g2) AS ?cases) WHERE {
      GRAPH ?g2 { ?c a cac:CACInvestigation }
      FILTER(STRSTARTS(STR(?g2), "https://w3id.org/caselinker/resource/case/")
         && !STRENDS(STR(?g2), "/provenance"))
  } }
} GROUP BY ?event ?cases
```

**Q6** [Cases with a victim under 12, of all cases and of cases with a recorded victim age]

```
SELECT ?n ?with_age ?cases
       (ROUND(1000 * ?n / ?cases) / 10 AS ?pct_of_cases)
       (ROUND(1000 * ?n / ?with_age) / 10 AS ?pct_of_cases_with_victim_age) WHERE {
  { SELECT (COUNT(DISTINCT ?g) AS ?n) WHERE {
      GRAPH ?g { ?person prod:victimAge ?age . FILTER(?age < 12) }
      FILTER(STRSTARTS(STR(?g), "https://w3id.org/caselinker/resource/case/")
         && !STRENDS(STR(?g), "/provenance"))
  } }
  { SELECT (COUNT(DISTINCT ?g) AS ?with_age) WHERE {
      GRAPH ?g { ?person prod:victimAge ?age }
      FILTER(STRSTARTS(STR(?g), "https://w3id.org/caselinker/resource/case/")
         && !STRENDS(STR(?g), "/provenance"))
  } }
  { SELECT (COUNT(DISTINCT ?g2) AS ?cases) WHERE {
      GRAPH ?g2 { ?c a cac:CACInvestigation }
      FILTER(STRSTARTS(STR(?g2), "https://w3id.org/caselinker/resource/case/")
         && !STRENDS(STR(?g2), "/provenance"))
  } }
}
```

**Q7** [Mean offender-victim age gap, over cases with both ages]

```
SELECT (COUNT(?g) AS ?cases_with_both) (ROUND(10 * AVG(?gap)) / 10 AS ?mean_gap_years) WHERE {
  { SELECT ?g ((AVG(?offender_age) - AVG(?victim_age)) AS ?gap) WHERE {
      GRAPH ?g {
        ?role a cac:OffenderRole ; core:hasRole ?p . ?p det:ageEstimate ?offender_age .
        ?q prod:victimAge ?victim_age
      }
      FILTER(STRSTARTS(STR(?g), "https://w3id.org/caselinker/resource/case/")
         && !STRENDS(STR(?g), "/provenance"))
  } GROUP BY ?g }
}
```

**Q8** [Cases with platform data, and with chat or Facebook]

```
SELECT ?platform (COUNT(DISTINCT ?g) AS ?n) ?cases
       (ROUND(1000 * COUNT(DISTINCT ?g) / ?cases) / 10 AS ?pct) WHERE {
  { GRAPH ?g { ?event cac:usesChannel ?p } BIND("any platform" AS ?platform) }
  UNION
  { GRAPH ?g { ?event cac:usesChannel <https://w3id.org/caselinker/resource/platform/chat> }
    BIND("chat" AS ?platform) }
  UNION
  { GRAPH ?g { ?event cac:usesChannel <https://w3id.org/caselinker/resource/platform/facebook> }
    BIND("facebook" AS ?platform) }
  FILTER(STRSTARTS(STR(?g), "https://w3id.org/caselinker/resource/case/")
         && !STRENDS(STR(?g), "/provenance"))
  { SELECT (COUNT(DISTINCT ?g2) AS ?cases) WHERE {
      GRAPH ?g2 { ?c a cac:CACInvestigation }
      FILTER(STRSTARTS(STR(?g2), "https://w3id.org/caselinker/resource/case/")
         && !STRENDS(STR(?g2), "/provenance"))
  } }
} GROUP BY ?platform ?cases
```

**Q9** [Cases with unknown or proactive investigation type]

```
SELECT ?type (COUNT(DISTINCT ?g) AS ?n) ?cases
       (ROUND(1000 * COUNT(DISTINCT ?g) / ?cases) / 10 AS ?pct) WHERE {
  { GRAPH ?g { ?case cac:investigationStatus "unknown" } BIND("unknown" AS ?type) }
  UNION
  { GRAPH ?g { ?case cac:hasStep ?op . ?op a tf:ProactiveOperation } BIND("proactive" AS ?type)
   }
  FILTER(STRSTARTS(STR(?g), "https://w3id.org/caselinker/resource/case/")
         && !STRENDS(STR(?g), "/provenance"))
  { SELECT (COUNT(DISTINCT ?g2) AS ?cases) WHERE {
      GRAPH ?g2 { ?c a cac:CACInvestigation }
      FILTER(STRSTARTS(STR(?g2), "https://w3id.org/caselinker/resource/case/")
         && !STRENDS(STR(?g2), "/provenance"))
  } }
} GROUP BY ?type ?cases
```

**Q10** [Methods that read the age 22 at characters 7,081 to 7,083 of nj_ag_2017_001 (Table 4)]

```
SELECT ?method ?family ?confidence ?start ?end ?quote WHERE {
  GRAPH <https://w3id.org/caselinker/resource/case/nj_ag_2017_001/provenance> {
    ?assertion a oa:Annotation ;
               dcterms:subject "age" ;
               prov:value ?value ;
               core:usesMethod ?method ;
               core:hasConfidence ?confidence ;
               oa:hasTarget ?target .
    ?target oa:hasSelector ?position , ?quote_sel .
    ?position oa:start ?start ; oa:end ?end .
    ?quote_sel oa:exact ?quote .
    ?method clprov:evidenceFamily ?family .
    FILTER(STR(?value) = "22" && ?start = 7081)
  }
} ORDER BY DESC(?confidence)
```

**Q11** [Resolution status by field, counted over provenance graphs]

```
SELECT ?field ?status (COUNT(DISTINCT ?g) AS ?cases) WHERE {
  GRAPH ?g {
    ?resolution a prov:Entity ;
                dcterms:subject ?field ;
                clprov:resolutionStatus ?status .
  }
  FILTER(STRENDS(STR(?g), "/provenance"))
} GROUP BY ?field ?status ORDER BY ?field DESC(?cases)
```

**Q12** [Case pairs that share a local agency and a location]

```
SELECT ?case_a ?case_b (COUNT(DISTINCT ?agency) AS ?shared_agencies)
                       (COUNT(DISTINCT ?location) AS ?shared_locations) WHERE {
  GRAPH ?ga { ?a mj:involvesAgency ?agency ; <https://w3id.org/caselinker/resource/vocab/
   mentionsLocation> ?location .
              ?agency a mj:LocalAgency }
  GRAPH ?gb { ?b mj:involvesAgency ?agency ; <https://w3id.org/caselinker/resource/vocab/
   mentionsLocation> ?location }
  FILTER(STR(?ga) < STR(?gb))
  FILTER(STRSTARTS(STR(?ga), "https://w3id.org/caselinker/resource/case/") && !STRENDS(STR(?ga),
    "/provenance"))
  FILTER(STRSTARTS(STR(?gb), "https://w3id.org/caselinker/resource/case/") && !STRENDS(STR(?gb),
    "/provenance"))
  BIND(?ga AS ?case_a)
  BIND(?gb AS ?case_b)
} GROUP BY ?case_a ?case_b ORDER BY DESC(?shared_locations) DESC(?shared_agencies) LIMIT 15
```